Ahnaf Rafi Sharan

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ResearchGate | LinkedIn

EDUCATION

Master of Science (M. Sc.) in Civil and Environmental Engineering — 11/2025 – Present
Islamic University of Technology (IUT) — Gazipur, Bangladesh
CGPA (Most Recent Semester): **3.75**/4.00

Bachelor of Science (B. Sc.) in Civil and Environmental Engineering — 06/2021 – 10/2025
Islamic University of Technology (IUT) — Gazipur, Bangladesh
CGPA: **3.91**/4.00 (Rank: 3rd out of 96)

RESEARCH INTERESTS

- Multimodal Traffic Data Analytics
- AI-Based Traffic Flow and Road Safety Analysis
- Digital Twins for Transportation Infrastructure
- Intelligent Transportation Systems (ITS)
- Transportation Policy and Planning
- Highway & Airport Pavement Engineering

RESEARCH EXPERIENCE

Developing Crash Prediction Model for Highways Considering Land Use and Encroachment (Undergraduate Thesis)
Supervisor: Moinul Hossain, Ph.D., Professor, CEE, IUT
- Compiled and processed crash data from the Dhaka–Sylhet National Highway (N2) and performed feature selection for crash prediction modeling.
- Developed and validated a Negative Binomial Regression model, with spatial and statistical visualization of crash-risk patterns.

Impact of unauthorized vehicles on road safety using machine learning in Bangladesh
Supervisor: Dr. Md. Rashedul Haque, Professor, Department of Civil Engineering, Pabna University of Science and Technology (PUST)
- Defined and categorized unauthorized vehicles in the Bangladesh context based on relevant legal and regulatory frameworks.
- Performed data sorting, cleaning, exploratory analysis, machine learning modeling, and visualization to investigate road safety implications.

Developing spatial crash prediction model for highways considering land use & encroachment
Supervisor: Moinul Hossain, Ph.D., Professor, CEE, IUT; Md. Asif Raihan, Ph.D., Professor, Accident Research Institute (ARI), BUET
- Applied QGIS-based geospatial analysis to examine crash interactions between adjacent highway segments and mapped roadside land use and encroachment patterns.
- Contributed to journal manuscript preparation and developed the research poster for presentation at the 105th TRB Annual Meeting (TRB 2026)

Understanding Crash Risk on Highways: The Impact of Land Use, Encroachment and Built Environment
Supervisor: Moinul Hossain, Ph.D., Professor, CEE, IUT; Md. Asif Raihan, Ph.D., Professor, ARI, BUET
- Investigated the interaction between land use, roadside encroachment, and built-environment characteristics and their influence on highway crash risk.
- Contributed to research manuscript development and designed the conference poster for presentation at the 16th EASTS Conference.

Utilization of recycled brick aggregate to ensure circular economy
Supervisor: Dr. Md. Tarek Uddin, PEng, Professor, CEE, IUT
- Developed predictive models for concrete compressive strength using Multivariable Linear Regression (MVLR) and Multivariable Full Quadratic (MVFQ) regression.
- Supported model development through statistical analysis and evaluation of influential concrete mix parameters.

Prediction of strength properties of soft soil considering simple soil parameters
Supervisor: Mozaher Ul Kabir Mahadi, Ph.D., Research Assistant, University of Kansas
- Compiled, sorted, and analyzed soft-soil data from 250 boreholes across Dhaka City to investigate relationships between basic soil parameters and strength properties.
- Prepared the complete journal manuscript and developed responses to peer-review comments and rebuttal requirements.

JOURNAL PUBLICATIONS

Haque, M. R., Hossain, M. N., **Sharan, A. R.**, Aman, N., Miah, A. S. M., Istiqphara, S., & Ferdous, M. F. (2026). *Impact of unauthorized vehicles on road safety using machine learning in Bangladesh.* **International Journal of Engineering**, 27/04/2026, e-ISSN 1735-9244.

Hoque, M. J., Bayezid, M., **Sharan, A. R.**, Kabir, M. U., & Tareque, T. (2023). *Prediction of strength properties of soft soil considering simple soil parameters*. **Open Journal of Civil Engineering**, *13*, 479–496. https://doi.org/10.4236/ojce.2023.133035

CONFERENCE PROCEEDINGS

Uddin, N., Tasnim, A., **Rafi, A.**, Raihan, M. A., Hossain, M., Fletcher, J., & Rahman, A. S. (2026). *Developing spatial crash prediction model for highways considering land use & encroachment*. Paper presented at **the 105th TRB Annual Meeting (TRB 2026)**, Washington, D.C., USA, January 11–15, 2026.

Uddin, N., Tasnim, A., **Rafi, A.**, Raihan, M. A., & Hossain, M. (2025). *Understanding Crash Risk on Highways: The Impact of Land Use, Encroachment and Built Environment*. Paper presented at **the 16th International Conference of Eastern Asia Society for Transportation Studies (EASTS 2025)**, Surakarta, Indonesia. https://doi.org/10.11175/easts.16.PP4087

Mohammed, T. U., Uddin, N., & **Rafi, A.** (2024). *Utilization of recycled brick aggregate to ensure circular economy*. In Proceedings of the 1st International Conference on Core Engineering & Technology (IUT-ICCET 2024), Gazipur, Bangladesh. **IUT Journal of Engineering and Technology (JET)**, *Special Issue*, 848–854.

PROFESSIONAL EXPERIENCE

Lecturer, Department of Civil Engineering 11/2025 – Present | Dhaka, Bangladesh
United International University (UIU)
Relevant Courses Taught:
- GIS and Remote Sensing Lab, Engineering Hydrology, Engineering Geology and Geomorphology

Industrial Training at SMEC (Bangladesh) Ltd. 03/2024 – 03/2024 | Dhaka, Bangladesh

ACADEMIC PROJECTS

Final Year Design Project (FYDP)
Design and Planning of an Eight-Story Residential Building in Kishoreganj
- Led the integrated design covering geotechnical, structural, environmental, and transportation aspects.
- Conducted mat and pile foundation analysis using PLAXIS 3D, selecting piles based on settlement and bearing capacity.
- Performed ETABS analysis and design of beams, columns, slabs, and stairs with serviceability and reinforcement checks.
- Conducted concrete mix design, service-life prediction, water demand analysis, and rainwater harvesting design using E-Tank.
- Conducted Traffic Impact Assessment and SUMO simulation to evaluate future traffic conditions.
- Prepared environmental assessment, EMP/EMoP, water and waste management plans, and pavement design.

RELEVANT COURSEWORK

- Transportation and Traffic Engineering
- GIS Application in Civil Engineering
- Transportation Projects and Operations
- Probability and Statistics
- Computer Programming and Applications
- Civil Engineering Data Analysis
- Public Transportation System
- Transportation Engineering Sessional I & II
- Numerical Methods and Computer Programming
- Pavement Design and Railway Engineering

SKILLS

Technical Skills
- **Analysis, Simulation and CAD Software:** AutoCAD, SUMO, VISSIM, CUBE, QGIS, ETABS, SAP2000, E-Tank, Zotero
- **Programming Language:** Python
- **Document Preparation:** Microsoft Office (Word, Excel and PowerPoint)

Language Proficiency
- **English:** Overall Band Score **7.5 (IELTS)**
- **Bengali:** Native

CERTIFICATIONS

- Certification of Accomplishment for completing Industrial Training at SMEC BD Ltd.
- Certification of Accomplishment for 'Data Visualization with Matplotlib' in Datacamp
- Certification of Accomplishment for 'Data Visualization with Seaborn' in Datacamp

EXTRACURRICULAR ACTIVITIES

CENNOVATION 2025 IUT, Gazipur
Hosted by Department of Civil and Environmental Engineering, IUT
Position: Assistant General Secretary

ACI Students' Chapters Meet 2025 IUT, Gazipur
Hosted by ACI IUT Student Chapter
Position: Design Team Lead

ACI IUT Student Chapter IUT, Gazipur
Position: Head of Creative Design

IUT ITE Student Chapter IUT, Gazipur
Position: Technical Executive

COMMUNITY INVOLVEMENT

Winter Clothes Donation, IUT SIKS 2023 – 2024 | Gazipur, Dhaka
Participated in donating winter clothing to underprivileged communities in Gazipur

Blood Donor, Badhan Jahangirnagar University Zone 2021 – Present | Savar, Dhaka
Active blood donor and volunteer in awareness campaigns